

Cosmetic Product Safety Assessment of

Glitter Gel / Hair Glitter in White , Pink and Purple color and Silver color

This safety assessment relates to the formulation described below. If the information below is incorrect, please amend and resubmit for reassessment.

KIMYON INDUSTRIAL (HK) COMPANY LTD
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838 LAI CHI KOK ROAD, KOWLOON, HK

PRODUCT FORMULATION

The chemical names shown below refer to the raw materials used to formulate this product. The identity of the raw materials is not necessarily based on the International Nomenclature of Cosmetic Ingredients (INCI) and does not represent the INCI listing that must be shown on the product label and is for assessment purposes only.

Chemical Name	Conc	% Active	Product	CAS No	Einecs No
AQUA (WATER)	90.2	100	90.2	7732-18-5	231-791-2
SORBITOL	3	100	3	50-70-4	200-061-5
PHENOXYETHANOL	0.9	100	.9	122-99-6	204-589-7
CARBOMER (BENZENE FREE)	0.5	100	.5	9007-20-9 / 9003-01-4 / 76050-42-5 / 9062-04-8 / 9007-16-3 / 9007-17-4	POLYMER
TRIETHANOLAMINE	0.3	100	.3	102-71-6	203-049-8
ETHYLHEXYLGLYCERIN	0.1	100	.1	70445-33-9	408-080-2
MAY CONTAIN (+/-)				-----	-----
POLYBUTYLENE TEREPHTHALATE	4.925	100	4.925	26062-94-2	POLYMER
POLYETHYLENE TEREPHTHALATE	4.92	100	4.92	25038-59-9	POLYMER
ACRYLATES COPOLYMER	0.075	100	.075	25133-97-5 / 25035-69-2 / 25212-88-8 / 25086-15-1 / 27322-15-2 / 9010-88-2	POLYMER
EPOXY RESIN	0.075	100	.075	61788-97-4	POLYMER
CI 77000 (ALUMINUM POWDER)	0.005	100	.005	7429-90-5	231-072-3
CI 17200 (D&C RED NO. 33)	0.07	100	.07	3567-66-6	222-656-9
CI 16035 (FD&C RED NO. 40)	0.1	100	.1	25956-17-6	247-368-0
CI 42090 (FD&C BLUE NO. 1)	0.03	100	.03	3844-45-9	223-339-8

LABELLED WARNINGS & INSTRUCTIONS OF USE

Keep away from eyes.

CONSUMER EXPOSURE

Product Class: Face and body glitter gel

IFRA Product type: Body Creams, Oils, Lotions, gels, paint of all types (except baby creams, lotions, oils)

IFRA Category: Category 4

Targeted Population: Children 16.7kg (Mean)

Amount per application/g: 3.910

Number of applications per day: Twice per day

Skin Surface Area of Application/cm²: 15670.000

Physical form: Gel

Total Amount applied per day/g: 7.82

Part of body exposed to undiluted

Skin of arms, body, face, hands, hair

Estimated Daily Exposure mg/kg/day: 130

Amount Per Unit Area of Skin per day mg/cm²/day: 0.510

Retention factor: 1.00

Exposure Time Neat: 720 Minutes

Exposure Time Dilute: Not Applicable

Exposure time Solvent Inhalation: Not Applicable

Exposure time Aerosol Inhalation: Not Applicable

This product has been assessed taking into account that it will be used on children aged five and above.

STABILITY OF COSMETIC PRODUCT

It is assumed that the responsible person has selected all pertinent criteria required to evaluate the stability of this cosmetic product during reasonable foreseeable conditions of storage. The stability report provided by the supplier and based upon the conclusions made therein, this cosmetic product appears to be stable under reasonably foreseeable storage conditions.

1-year stability test data at -18°C, 5°C and 40 °C as well as cycle test (2020.12.31) has been provided by applicant. Based on the physiochemical data as well as microbiological data stated, the product is stable in the finished packaging for the tested period.

MICROBIOLOGICAL QUALITY

To comply with the Guidelines on the Microbiological Quality (SCCS/1564/15), the following maximum limits apply:

Category 1: Products specifically intended for children under 3 years, eye area and mucous membranes.

TVC:- 100 cfu/g or ml in 0.5 g or ml of the product.

Pseudomonas aeruginosa, Staphylococcus aureus and Candida albicans must not be detectable in 0.5 g or ml of the cosmetic product

Category 2: Other cosmetic products.

TVC:- 1000 cfu/g or ml in 0.1 g or ml of the product

Pseudomonas aeruginosa, Staphylococcus aureus and Candida albicans must not be detectable in 0.1 g or ml of the cosmetic product

The microbiological specifications for the product have been supplied and based upon the conclusions therein, meet the Industry requirements specified in the SCCS's Notes of Guidance for the Testing of Cosmetic Ingredients and their Safety Evaluation 8th Revision, Section 4-4.2

The preservative challenge test results for this product have been supplied and based upon the conclusions made therein appear to meet the industry requirements specified in the SCCS's Notes of Guidance for the Testing of Cosmetic Ingredients and their Safety Evaluation 8th Revision, Section 4-4.3

Microbiological contamination test report (2021.09.10) has been provided by applicant, demonstrating hygiene condition of product complying to limits under Cat. 2 stated above.

Preservative challenge test report (2021.08.17) has been provided by applicant, demonstrating performance of product formulation against contamination from index micro-organisms.

PACKAGING INFORMATION/COMPATIBILITY

It is assumed that the responsible person has identified the most applicable testing required to determine the packaging stability and its interaction with the cosmetic product contained within it. Taking into consideration the information supplied to the assessor, there appears to be no immediate health concern due to the characteristics of packaging materials in direct contact with the final product.

Testing data on plastic container have been provided (2019.06.25), indicating the components meeting the specification of overall and specific migration on acrylonitrile/primary aromatic amine for food packaging as per Regulation (EC) No. 1935/2004 and Regulation (EU) No. 10/2011 on materials and articles intended to come into contact with food. In addition, testing data on plastic components have been provided (2019.06.25), indicating the components not containing heavy metals at levels of concern, with lead, cadmium, mercury and chromium (VI) below 100 ppm requirement as laid down under Directive 94/62/EC for packaging and packaging waste.

Testing data on metal component have been provided (2019.06.25), indicating the release of nickel content from the container meeting the requirement under Annex XVII Item 27 of the REACH Regulation (EC) No. 1907/2006 & Amendment No. 552/2009.

SERIOUS / UNDESIRABLE EFFECTS

The supplier has confirmed that no undesirable effects or serious undesirable effects of this cosmetic product have been reported or, where relevant, cosmetic products with a similar formulation.

HUMAN STUDIES

No existing studies from human volunteers were provided at the time of assessment.

FRAGRANCE COMPOSITIONS

This formulation does not contain a synthetic fragrance and therefore a fragrance safety evaluation as per IFRA code of practice is not applicable to this product.

PRESENCE OF NANOMATERIALS

The supplier has confirmed that this cosmetic product does not contain any nanomaterials that are known to them within the meaning of the definition as stated in Cosmetic Product Regulation (EC) No 1223/2009.

IMPURITIES /TRACES/ PROHIBITED SUBSTANCES

Where the specification is provided it is noted that this product does not contain any impurities at levels likely to cause harm to the user

TOXICOLOGICAL & REGULATORY REVIEW

An aqueous mixture of emollient, preservative, viscosity controlling agent, buffer, skin conditioning agent, glitter and colorant, intended to apply on face, body and hair by children 5 years and up for beautifying purpose. Repeated exposure to hair and scalp, face and body is expected. Accidental exposure to eyes during normal use is anticipated but not expected to be a systemic one. The preservative and colorants are approved for use in cosmetics under EU cosmetic requirements and within the restrictions of use stated if any, while none of the ingredients is on the prohibition list of the regulation. The ingredient of triethanolamine is listed in the restricted substance annex (Annex III/62). The grade of triethanolamine is reported as 99% purity with less than 0.5% of diethanolamine and less than 20 microgram/kg of nitrosamine, which complies with the requirements of the Cosmetics Regulation. Cross reactions among the ingredients are unlikely to happen based on a review of the chemical compositions of the ingredients. Most of the ingredients being used are not acutely oral or dermal toxic. Evaluation of the safety of specific concerned ingredients is stated as follows:

Phenoxyethanol is irritating in nature and is a common preservative used in cosmetic. The current use level meets the restriction under Annex V for approved preservatives. With a NOAEL of 300 mg/kg bw/day determined from developmental toxicity dermal study in rabbits, an MoS of 100 has been determined, being the lowest in this product formulation.

No NOAEL can/has been determined for polyethylene terephthalate, polybutylene terephthalate and epoxy resin as of today. They have been used as glitter, which is of high molecular weight and, taking into account of the physical form of the product also, absorption through skin and digestive system is expected to be minimal. Other than potential mechanical irritation of the glitter, taking into account of their long history of safe use in the cosmetic industry, risk aroused from the use in the current product formulation should be low.

Where a NOAEL is available for a chemical ingredient that is considered as a toxicological concern, the Margin of Safety (MoS) has been calculated as greater than 100 taking into consideration any known data on dermal absorption and bioavailability. It is generally accepted that the MoS should be a least 100 to declare a substance safe for use in a finished product and the safety of this formulation is further supported by this uncertainty factor.

If used as directed, use of this product should be uneventful.

Health effects of the product as supplied on the skin

The formulation as supplied may cause slight skin irritation especially if exposure is prolonged and/or repeated.

Repeated exposure to the formulation as supplied is unlikely to produce allergy by skin contact.

Exposure to this product is unlikely to result in phototoxic effects.

Unlikely to cause damage to internal organs following absorption through the skin.

Health effects of the product as supplied on the eye

Accidental exposure of the eye to the formulation as supplied may result in slight eye irritation.

Health effects following ingestion of the product as supplied

The formulation as supplied if swallowed may cause slight, transient irritation to the mouth and upper digestive tract.

Health effects of inhaling the product

Inhalation is an unlikely route of exposure

Overall Assessment Conclusion

The ingredients are legally permitted as per Cosmetic Regulation (EC) No. 1223/2009 and its amendments and the safety assessment has been carried out in accordance to this regulation. They must comply with the relevant purity standards for cosmetic ingredients. It is assumed that these ingredients do not contain any undisclosed impurities/contaminants that would affect the conclusions reached. The product must be manufactured in accordance with EU Guidance on Good Manufacturing Practice.

Keep away from eyes.

Under normal or reasonably foreseeable conditions of use, a product made to this formulation is unlikely to produce an abnormally high number of adverse reactions. The product will give users the level of safety they can reasonably expect when used as directed.



Cosmetic Regulations Product Safety Assessor

Suzanna Chan BSc, MSc Appl Tox, PhD, CChem, FRSC, UKRT, ERT

2/F Garment Centre, 576 Castle Peak Road, Kowloon, Hong Kong SAR, China **Date:** 05 Oct 2021

Glitter Gel / Hair Glitter in White , Pink and Purple color and Silver color

TOXICOLOGICAL PROFILE OF SUBSTANCES

Chemical Substance: AQUA (WATER)

EU INCI NAME:AQUA

CAS: 7732-18-5

EINECS 231-791-2

Appearance: Liquid

Water Solubility: highly soluble

Function: Solvent

Melting Point: 0

Boiling Point: 100

Cosmetic Regulatory Summary:

EU Cosmetics Status: Not controlled
ASEAN Cosmetics Status: Not controlled
Saudi Cosmetics Status: Not controlled by Saudi legislation
US Cosmetics Status: Not controlled
Canadian Cosmetics Status: Not controlled

Regulatory Summary:

EU Classification: Unclassified
GHS Classification: Unclassified
REACH Annex XVII controlled: Not Controlled
REACH SVHC Candidate List: Not Controlled

Systemic Exposure Dosage / Margin of Safety:

SED Adult mg/kg bw/day: 117.560	No NOAEL Available	NO(A)EL mg/kg bw day: -
SED Child mg/kg bw/day: 422.373	No NOAEL Available	NOAEL test method: -
SED Baby mg/kg bw/day: 1195.532	No NOAEL Available	

Toxicological Summary:

Function : Solvent. Simply water unlikely to cause irritation, allergy or harm. Used in many cosmetic products as a solvent and necessary to sustain biological life. The source of water should be known, monitored to GMP and either a deionised or high purity grade free from toxins, pollutants and bacteriological contamination should be used in cosmetic products.

Chemical Substance: SORBITOL

EU INCI NAME:SORBITOL

CAS: 50-70-4

EINECS 200-061-5

Appearance: Crystalline powder

Log Kow: Not determined

Water Solubility: Soluble (up to 83%)

Function: Humectants / Emulsifier / Flavouring agent

Melting Point: 90 -112°C

Boiling Point: 295°C

Vapour Pressure: 3.1kPa (35°C)

Cosmetic Regulatory Summary:

EU Cosmetics Status: Not controlled
ASEAN Cosmetics Status: Not controlled
Saudi Cosmetics Status: Not controlled by Saudi legislation
US Cosmetics Status: Not controlled
Canadian Cosmetics Status: Not controlled

Regulatory Summary:

EU Classification: Not classified
GHS Classification: Not classified
REACH Annex XVII controlled: Not Controlled
REACH SVHC Candidate List: Not Controlled

Systemic Exposure Dosage / Margin of Safety:

SED Adult mg/kg bw/day: 3.128	MoS - Adult 60kg: 2397.6	NO(A)EL mg/kg bw day: 7500
SED Child mg/kg bw/day: 11.238	MoS - Child 16.7kg: 667.3	NOAEL test method: 2-year dietary study; rat
SED Baby mg/kg bw/day: 31.810	MoS - Baby 5.9kg: 235.7	

Toxicological Summary:

Cosmetic Functions : Humectant / Plasticiser /Skin Conditioning / Flavoring Agent / Fragrance Ingredient. Sorbitol is a hexahydric sugar alcohol that occurs naturally in plants and animals. It is widely used in the food industry as an ingredient in many food products and medically, sorbitol is used in parenteral nutrition as a source of carbohydrate. It has long history of use as food ingredient/additive and there have been no indications of significant harmful effects reported. Sorbitol is hygroscopic. About 80% of an orally administered dose is absorbed. Once absorbed, it is rapidly metabolised by the liver to several intermediates, but ultimately to glucose. Sorbitol is well utilised by the body as an energy source, and consequently, very little of an oral dose is eliminated in the urine and faeces. The only consistent adverse effect found in animals or humans is mild gastrointestinal effects (GI) such as gas, bloating, abdominal discomfort, and diarrhoea.

Acute toxicity: Sorbitol was practically non-toxic in acute oral studies in laboratory animals (LD50: 7.1 to 25 g/kg bw) and short-term toxicity studies in humans demonstrates that it does not cause any adverse effect other than mild gastrointestinal effects even at high dose levels of between 10-50 g.

Irritation: Non-irritating. There is no data available indicating that sorbitol has skin, eye (mucous membrane) or respiratory tract irritation potential.

Sensitisation: Non-sensitising. No data available in animals and sorbitol is rarely associated with elicitation of allergic dermatitis in humans including occupational exposure.

Mutagenicity/Genotoxicity: Sorbitol is not genotoxic in a wide range of in vitro and in vivo mutagenicity tests.

Repeat dose toxicity: No toxic effects were noted in rats when 5% sorbitol was given in the diet for 3 months corresponding to 5 g/kg bw/day. Administered intravenously to rabbits, no effect on blood chemistry, body weight, survival, or organs at doses of 2.5, 5.0, or 10 mL/kg of a 20% sorbitol solution (500 mg/kg bw/d, 1000 mg/kg bw/d, and 2000 mg/kg bw/d) given daily for 3 months. Sorbitol was fed to weanling rats for two years at dietary levels of 0, 1, 5, or 10%. The two highest dose levels were supplemented with casein to maintain a 25% dietary protein level. A few of the treated animals had diarrhoea during the first 2 months at the 10% feeding level (12 g/kg bw/day). No toxicological effects were reported; the NOAEL was > 12000 mg/kg bw/d. In another 2-year chronic toxicity study, rat fed 20% sorbitol in the diet (~7.5 g/kg bw/d) had enlarged cecum, lower body weight gain, absolute thyroid weight and increased incidence of adrenal medullary hyperplasia. There is no evidence for reproduction, teratogenicity or developmental toxicity potential shown with sorbitol in numerous studies conducted in laboratory animals even at high doses of 1600 – 7800 mg/kg bw/d. Treatment-related tumours were not reported in the chronic toxicity studies which indicates that sorbitol lacks carcinogenicity activity.

Photo-induced toxicity: Non-phototoxic.

Other data (human): Sorbitol causes laxation because of its osmotic effects and poor absorption by the small intestine which results in increased transit time through the colon. In humans, the laxative threshold is 10-20 g of sorbitol liquid and 50 g of crystalline sorbitol; a study in 86 volunteers at a dose level of 25 g/day in two doses did not cause laxation. According to the results of dietary survey conducted by the US National Research Council (NRC), the average daily dietary intake of sorbitol ranges from 770 mg/day in babies to ~31g/day in adults or a maximum of 55.4 g/day.

Dermal / percutaneous absorption: No data available. However, it is assumed that the absorption through the skin would be lower than that of the oral routes; thus, the oral absorption value of 80% is used as conservative estimate.

EU INCI NAME:PHENOXYETHANOL

CAS: 122-99-6

EINECS 204-589-7

Function: preservatives

Appearance: Liquid

Log Kow: 1.16

Water Solubility: 28.9g/L

Melting Point: N/A

Boiling Point: Data not available.

Vapour Pressure: Data not available.

Cosmetic Regulatory Summary:

EU Cosmetics Status: Approved preservative
 ASEAN Cosmetics Status: Approved preservative - Maximum conc. 1% (Annex VI/#29)
 Saudi Cosmetics Status: Permitted preservative all products. Max conc 1% by weight .
 US Cosmetics Status: Up to 5% (CIR, 2009).
 Canadian Cosmetics Status: Not controlled

Regulatory Summary:

EU Classification: R22-36
 GHS Classification: H319- Eye Irrit. 2
 H302- Acute Tox. 4
 REACH Annex XVII controlled: Not Controlled
 REACH SVHC Candidate List: Not Controlled

Systemic Exposure Dosage / Margin of Safety:

SED Adult mg/kg bw/day: 0.997	MoS - Adult 60kg: 358.0	NO(A)EL mg/kg bw day: 357	
SED Child mg/kg bw/day: 3.582	MoS - Child 16.7kg: 99.6	NOAEL test method:	Sub-chronic (90 days) toxicity (dermal)
SED Baby mg/kg bw/day: 10.139	MoS - Baby 5.9kg: 35.2		

Toxicological Summary:

Phenoxyethanol is an aromatic ether with broad range of anti-microbial activity used in cosmetics as a preservative at concentrations below 1% and as a fixative for perfumes. Phenoxyethanol may contain free phenol (less than 1%) or the diethoxylate as impurities; cosmetic-grade Phenoxyethanol is generally 98% and is free of ethylene oxide (< 1 ppm) (CIR, 1990 and references cited therein). Listed in Annex VI/29 of Regulation (EC) 1223/2009 as approved preservative; maximum concentration in ready for use preparation – 1%. ¹⁴C-Phenoxyethanol was rapidly and almost completely absorbed from the gastrointestinal tract with no bioaccumulation potential detected. 89-93% of the absorbed was excreted in the urine with the majority occurring within the 6 hours; about 2% of administered dose was eliminated in the faeces. The main metabolite detected in the urine was phenoxyacetic acid via the oxidation of the terminal hydroxyl group to a carboxylic acid. Other metabolites have been identified as the hydroxylated, sulfated and/or glucuronidated products.

Acute toxicity: Phenoxyethanol was essentially nontoxic on oral, dermal or inhalation exposure in laboratory animals. The oral LD50 was 1850 mg/kg bw/d (rat) and > 5000 mg/kg bw in acute dermal study in rabbits (REACH Dossier). The dermal LD50 in rat for undiluted Phenoxyethanol was 13.0 (range 10.3-15.4) ml/kg. The LC50 was 1000 mg/m³ (rat).

Irritation: Non-irritating to skin under occlusive condition in test on rabbits (mean erythema score: 1; edema = 0). When Phenoxyethanol was applied daily to the shaved backs of guinea pigs for a total of ten applications, it was noted that all of the test animals had slight erythema, which was not worsened with the repeated applications. Phenoxyethanol was irritating to eyes in rabbit test inducing slight cornea opacity, moderate to severe conjunctivae erythema and chemosis which were all reversible within 15 days post-exposure.

Sensitisation: Not sensitizing in guinea pig maximization test with challenge dose of up to 10% concentration.

Mutagenicity/Genotoxicity: Phenoxyethanol was non-mutagenic in the Ames test (with and without metabolic activation) and in the mouse micronucleus test (CIR, 1990).

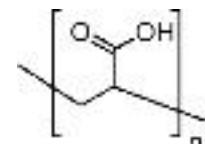
Repeat dose toxicity: Phenoxyethanol was applied dermally to rabbits at a dose of 0, 50, 150, or 500 mg/kg bw/day for 13 weeks; no mortality or overt signs of toxicity was noted. There was no effect on haematology, clinical chemistry, organ weights or any significant necropsy and microscopic findings. The only treatment related findings was observation of erythema and slight scaling of the skin at the site of treatment, NOAEL was concluded to be 500 mg/kg bw/d (REACH Dossier). Sub-chronic dietary administration of phenoxyethanol at dose levels of 500, 2500 or 10000 ppm (ca. 34, 169 or 697 mg/kg bw/d -males; 50, 234, or 939 mg/kg bw/d –females) for 13 weeks followed by a recovery period of 4 weeks showed no remarkable treatment related effects on body weights, haematological or clinical chemistry parameters. Slight increase in some organ weights (liver, kidney, spleen and seminal vesicles) were reported at the highest dose; however in the absence of any associated significant histopathological findings, were considered to be toxicologically irrelevant. The NOAEL in the study was concluded as the top dose of 10000 ppm (ca. 700 mg/kg bw/d) (REACH Dossier). In a sub-acute study in dogs dosed oral capsule of phenoxyethanol at doses of 500 or 1000 mg/kg bw/d, clinical signs of toxicity including vomiting, salivation and unsteady gait occurred at the top dose; NOAEL was 500 mg/kg bw/d. 14 days nose-only inhalation study with phenoxyethanol at concentration of 0, 40, 200 or 1000 mg/m³ resulted in histopathological changes in the upper respiratory tract and increased lung weights with the identified NOAEC of 48.2 mg/m³. A 2-generation feeding study to determine the reproductive and fertility effect of phenoxyethanol was conducted in mice administered 0, 0.25%, 1.25% or 2.5% in the diet; effect on body weight of paternal and neonatal animals was observed and increase in liver weight of F0 and F1 generation. No effect on reproductive parameter, treatment-related malformations or variation was observed. The NOAEL for reproduction was concluded as 1875 mg/kg bw/d (ca.) and 375 mg/kg bw/d (ca.) for foetal toxicity (REACH Dossier). Developmental toxicity or teratogenicity potential was not observed in rats dosed orally with 100, 300 or 1000 mg/kg bw/d phenoxyethanol on gestation day 6 to 15 and in dermal application study in pregnant rabbits at levels of 300, 600 or 1000 mg/kg bw/d; NOAEL of 300 mg/kg bw/d was identified based on maternal toxicity in both studies (CIR, 1990; REACH Dossier). Sub-chronic (90 days) toxicity (dermal): Based on the lack of treatment-related effects on body weight, organ weights, haematological and clinical chemistries and gross and histopathological examinations, the no-observed-adverse-effect level (NOAEL) for systemic toxicity was concluded to be 500 mg/kg bw/day under the conditions of this study. To account for the dosing schedule used in this study, the SCCS adjusted the NOAEL which should be multiplied by a factor of 5/7 to give an adjusted NOAEL of 357 mg/kg bw/day (SCCS Opinion SCCS/1575/16).

Photo-induced toxicity: Phenoxyethanol was not phototoxic in clinical studies

Other data (human): Phenoxyethanol, 10% (v/v) in mineral oil was tested for irritation and sensitization potential in a repeated insult patch test using a panel of 51 subjects, no cutaneous reaction was noted and it was concluded to be neither a primary nor a cumulative irritant and showed no evidence of delayed contact hypersensitivity. Furthermore, of 2736 patients patch-tested with 1% Phenoxyethanol in petrolatum none had signs of irritant or allergic reactions 2 and 4 days after application. A positive reaction rate of 0.2% has been reported in routine patch testing of 501 patients for suspected contact dermatitis with 5% Phenoxyethanol applied in petrolatum. Allergic contact dermatitis to 1% Phenoxyethanol could be a rare possibility in patients having an adverse reaction to aqueous creams as shown in one case report (CIR, 1990).

Dermal/percutaneous absorption: The percutaneous penetration of phenoxyethanol has been investigated in vitro using rat (non- occluded and occluded) and human donor skin (Roper et al. 1997). The permeability and amount absorbed were similar for both human and rat at approximately 60%.

Margin of Safety: Dermal toxicity study: NOAEL: 357 mg/kg bw/d; Dermal Absorption (Dap) = 47% rinse off, 85% leave on.



Chemical Structure:

EU INCI NAME: CARBOMER

CAS: 9007-20-9 / 9003-01-4 / 76050-42-5 / 9062-04-8 / 9007-16-3 / 9007-17-4

Function: Emulsion stabilising/Gel forming/Viscosity controlling

EINECS polymer

Appearance: solid

Cosmetic Regulatory Summary:

EU Cosmetics Status: Not controlled
ASEAN Cosmetics Status: Not controlled
Saudi Cosmetics Status: Not controlled by Saudi legislation
US Cosmetics Status: Safe for use up to 2% typically (CIR 2009)
Canadian Cosmetics Status: Not controlled

Regulatory Summary:

EU Classification: Unclassified
GHS Classification: Unclassified
REACH Annex XVII controlled: Not Controlled
REACH SVHC Candidate List: Not Controlled

Systemic Exposure Dosage / Margin of Safety:

SED Adult mg/kg bw/day: 0.651	MoS - Adult 60kg: 1743.2	NO(A)EL mg/kg bw day: 1136	
SED Child mg/kg bw/day: 2.341	MoS - Child 16.7kg: 485.1	NOAEL test method:	rat, repeated dose toxicity oral feed, 28 d study
SED Baby mg/kg bw/day: 6.627	MoS - Baby 5.9kg: 171.4		

Toxicological Summary:

A polymeric thickening agent also used as film formers and binding agents. Polyacrylic acids have a low potential to cause irritation or allergy. The Cosmetic Ingredients Review Panel reaffirmed its conclusion that carbomers showed low potential for skin irritation and sensitization at concentrations up to 100%. It also referenced that Carbomers have low toxicity if ingested. Some grades of Carbomer contain benzene as a result of the manufacturing process. The EU Cosmetics Directive prohibits the use of benzene except for traces that are technically unavoidable or cannot be reasonably removed in Good Manufacturing Practice and provided such traces cannot cause harm to human health. As there are benzene-free grades that are available, the grade used must not contain benzene in order to comply with the EU Cosmetic Product Directive. This particular grade is of a benzene-free grade and considered as a suitable and acceptable purity for use in cosmetic products. Carbomers are considered as a safe cosmetic ingredient in present practices of use. A NOAEL (rat, oral feed), 28 d repeated dose toxicity study of 1,136 mg/kg bw/d was established (HERA project 2009: Polycarboxylates used in detergents (Unilever 1993).

Chemical Substance: TRIETHANOLAMINE

EU INCI NAME: TRIETHANOLAMINE

CAS: 102-71-6

Function: Buffering / Emulsifying / Masking / Surfactant

EINECS 203-049-8

Appearance: liquid

Melting Point: 20.5 °C

Log Kow: Not available.

Boiling Point: 335.4 °C

Water Solubility: Miscible

Vapour Pressure: Not available.

Cosmetic Regulatory Summary:

EU Cosmetics Status: Controlled (2.5% max - leave-on); Avoid Nitrosating agents
ASEAN Cosmetics Status: Controlled (2.5% max - leave-on); Avoid Nitrosating agents
Saudi Cosmetics Status: Controlled (2.5% max - leave-on); Avoid Nitrosating agents
US Cosmetics Status: safe for use in rinse-off products; 5% leave-on (CIR)
Avoid Nitrosating agents

Regulatory Summary:

EU Classification: R36, R37, R38
GHS Classification: Unclassified
REACH Annex XVII controlled: Not Controlled
REACH SVHC Candidate List: Not Controlled

Systemic Exposure Dosage / Margin of Safety:

SED Adult mg/kg bw/day: 0.391	MoS - Adult 60kg: 2557.5	NO(A)EL mg/kg bw day: 1000	
SED Child mg/kg bw/day: 1.404	MoS - Child 16.7kg: 711.8	NOAEL test method:	Oral, rat, repeated dose
SED Baby mg/kg bw/day: 3.976	MoS - Baby 5.9kg: 251.4		

Toxicological Summary:

The grade of Triethanolamine must be one of minimum 99% purity and less than 0.5% diethanolamine in order to comply with the requirements of the EU Cosmetics Regulation; Annex III/62. Where specification is not supplied at assessment, it is assumed that the supplier will be using raw materials in compliance with those requirements. If this is not the case the supplier must bring this to the attention of the assessor immediately. A base soluble in water and organic solvents. In free base form irritant to skin and eye, when ionised has lower potential to cause irritation. Used in cosmetics to neutralise polymers or to form soaps. Mild skin and eye irritants leading to restriction in prolonged contact with skin at higher concentrations but little skin sensitisation in clinical testing at the typical low concentrations. Non mutagenic in the Ames test, and to Bacillus subtilis. No carcinogenic activity was reported through dermal data. Single-dose oral toxicity (rat) - practically non toxic to slightly toxic, little potential for rabbit skin irritation in acute and subchronic skin irritation tests. The NOEL for triethanolamine is 1125mg/kg/day (developmental toxicity - ORAL GAVAGE 6-15 DAY) where as for oral the NOAEL is (1000 mg/kg/day) the lowest NOAEL. No carcinogenic or cocarcinogenic activity when dermally applied to mice for 18 months. Clinical skin testing of TEA and cosmetic products containing TEA and DEA showed mild skin irritation in concentrations above 5 percent. There was very little skin sensitization. There was no phototoxicity or photosensitization reactions with products containing up to 20.04 percent TEA. Some reports in Contact Dermatitis of allergenic reaction in consumer products. Data above taken from CIR Compendium 2011. In the event no MSDS or purity specification was provided to the safety assessor, the supplier must check the grade of triethanolamine to ensure it meets the purity requirements of the EU Cosmetics Directive before cosmetic products containing this ingredient enters the marketplace. Do not use with nitrosating systems such as bronopol. Maximum nitrosamine content: 50 µg/kg (applies to finished product). Keep in nitrite free containers. Maximum limit of 2.5% triethanolamine in leave-on products. No limit in rinse-off products. CIR concludes safe for use in rinse-off products; but up to 5% in leave-on cosmetic products. Although no harmonized classification is recorded under CLP, it has been notified by most manufacturers in C&L Inventory as "Eye Irrit. 2 H319: Causes serious eye irritation".

References:

1. CIR Compendium 2011
2. Hamilton TK, Zug KA (1996). "Triethanolamine allergy inadvertently discovered from a fluorescent marking pen". Am J Contact Dermat 7 (3): 164–5
3. IUCLID datasheet
4. Allergic contact dermatitis from triethanolamine in a sunscreen". Contact Dermatitis 44 (1): 41–2.

EU INCI NAME:ethylhexylglycerin

CAS: 70445-33-9

EINECS 408-080-2

Appearance: Colorless to pale yellow viscous liquid

Log Kow: 2.53

Water Solubility: 1.8

Function: Masking/Skin Conditioning

Melting Point: N/A

Boiling Point: >285

Vapour Pressure: 25

Cosmetic Regulatory Summary:

EU Cosmetics Status: Not controlled

Saudi Cosmetics Status: Not controlled by Saudi legislation

US Cosmetics Status: Not controlled

Canadian Cosmetics Status: Not controlled

Regulatory Summary:

EU Classification: Xn; Xi; R20; R41-52/53

GHS Classification: Eye Dam. 1, Aquatic Chronic 3.

REACH Annex XVII controlled: Not Controlled

REACH SVHC Candidate List: Not Controlled

Systemic Exposure Dosage / Margin of Safety:

SED Adult mg/kg bw/day: 0.071 MoS - Adult 60kg: 697.5

SED Child mg/kg bw/day: 0.257 MoS - Child 16.7kg: 194.1

SED Baby mg/kg bw/day: 0.728 MoS - Baby 5.9kg: 68.5

NO(A)EL mg/kg bw day: 50

NOAEL test method: 90-day study daily by oral gavage at dosages of 50, 200 and 800 mg/kg/day in rats

Toxicological Summary:

Ethylhexylglycerin is known also as Octoxyglycerin. As supplied classified as severely irritating to eyes but a 5% solution in water is said to be non irritating to eyes. It is reported that this may be the first case of contact dermatitis to Ethylhexylglycerin reported in a 57-year-old woman (Linsen G and Goossens A. 2002. *Allergic contact dermatitis from ethylhexylglycerin. Contact Dermatitis*, 47:165). A positive (+) reaction to ethylhexylglycerin (10% pet.) on days 2 and 4 was obtained following patch testing. Five controls tested negatively. The concentration tested was not stated (Linsen G and Goossens A. 2002). Its use should be carefully monitored although use at up to 5% in a cosmetic product is described as unlikely to cause irritation or allergy when used in cosmetic products. Other subsequent allergic contact dermatitis (ACD) to Ethylhexylglycerin have been reported in 64 and 68 year old woman respectively (Stausbol-Gron, B and Andersen, K E. 2007. *Allergic contact dermatitis to ethylhexylglycerin in a cream. Contact Dermatitis*, 57:193-194; Mortz, CG *et al.* 2009. *Allergic contact dermatitis to ethylhexylglycerin and Pentylene Glycol. Contact Dermatitis*, 61, 180). Allergic Contact Dermatitis developed in these women from using hand and facial creams containing Ethylhexylglycerin. A 5% Ethylhexylglycerin in petroleum produced a strong positive patch test reaction (++) (Stausbol-Gron, B and Andersen, K E. 2007; Mortz, CG *et al.* 2009). This ingredient has a preservatives properties although it is not listed as a preservative in the EU cosmetics Directive 76/768/EEC as amended nor in the Regulation (EC) No. 1223/2009. Its main property is reported to be enhancing the uptake of preservatives by micro-organisms, thus potentiating the efficacy of preservatives (Stausbol-Gron, B and Andersen, K E. 2007). Allergic contact dermatitis to ethylhexylglycerin in a cream. *Contact Dermatitis*, 57:193-194). In another sensitization study, among 557 patients tested with Ethylhexylglycerin in the Contact Allergy Unit in Leuven, Belgium, 10 were found to be sensitized (Contact Dermatitis 74(5): 281-288). An additional 3 cases sensitized to Ethylhexylglycerin were identified in Antwerp. The oral and dermal LD50 is >2000 mg/kg. The LC50 for male and female rats is 2.83 and 3.22 mg/L respectively. Mild irritation was observed in rabbits applied with ethyl hexyl glycerin. Severe eye irritation was observed in rabbits. Mild eye irritation was exhibited with 5% solution. Moderate to severe conjunctival irritation was observed with the neat chemical. Non sensitizer to guinea pigs. Non mutagenic and non-clastogenic effect were observed in Ames bacterial reverse mutation assay and bone marrow micronucleus assay respectively. In a repeated dose oral toxicity study for a period of 90 days in rats showed increased incidence of renal mineralisation, hepatic hypertrophy at the high dose (800 mg/kg), and increased liver weight in all doses. Therefore the low observed adverse effect level (LOAEL) was derived to be 50 mg/kg/day. In further study report from submitted at CIR EXPERT PANEL MEETING SEPTEMBER 26-27, 2011, a NOAEL of 50mg/kg bw/day was recorded on the basis that the liver effects were not dose related and reversible. Ethylhexylglycerin is described as a rare contact allergen but its potential to cause ACD may be overlooked if not specifically tested in those individuals showing evidence of dermatitis from using cosmetic products. Specifically prone to develop new contact allergies are people with multiple contact allergies, chronic relapsing dermatitis, those with prolonged and repeated topical treatments or personal predisposition(Stausbol-Gron, B and Andersen, K E. 2007).

Chemical Substance: MAY CONTAIN (+/-)

EU INCI NAME:-

CAS: -----

EINECS -----

Appearance: -

Log Kow: -

Water Solubility: -

Function: -----

Melting Point: -

Boiling Point: -

Vapour Pressure: -

Cosmetic Regulatory Summary:

EU Cosmetics Status: -----

ASEAN Cosmetics Status: -----

Saudi Cosmetics Status: -----

US Cosmetics Status: -----

Canadian Cosmetics Status: -----

Regulatory Summary:

EU Classification: -----

GHS Classification: -----

REACH Annex XVII controlled: -----

REACH SVHC Candidate List: -----

Systemic Exposure Dosage / Margin of Safety:

SED Adult mg/kg bw/day: .000 No NOAEL Available

SED Child mg/kg bw/day: .000 No NOAEL Available

SED Baby mg/kg bw/day: .000 No NOAEL Available

NOAEL test method:

Toxicological Summary:

Chemical Substance: POLYBUTYLENE TEREPHTHALATE

EU INCI NAME:POLYBUTYLENE TEREPHTHALATE

CAS: 26062-94-2

EINECS polymer

Function: Film-former / viscosity controller

Cosmetic Regulatory Summary:

EU Cosmetics Status: Not controlled

US Cosmetics Status: Not controlled

Canadian Cosmetics Status: Not controlled

Regulatory Summary:

EU Classification: Unclassified

GHS Classification: Unclassified

REACH Annex XVII controlled: Not Controlled

REACH SVHC Candidate List: Not Controlled

Systemic Exposure Dosage / Margin of Safety:

SED Adult mg/kg bw/day: 6.418 No NOAEL Available

SED Child mg/kg bw/day: 23.061 No NOAEL Available

SED Baby mg/kg bw/day: 65.277 No NOAEL Available

Toxicological Summary:

Cosmetic functions: film former; hair fixative and viscosity increasing agent. Polybutylene terephthalate is a copolymer of 1,4-butanediol and dimethyl terephthalate or terephthalic acid. Inert fine particles which reflect the light. Not irritating to skin but may cause a foreign body reaction in the eye which could enhance the eye effects of other components in the formulation. Will cause a foreign body reaction if inhaled. Polybutylene terephthalate is approved for use as a substance that may come in contact with food in packaging (21CFR177.1660).

CIR: safe as used (up to 12%).

As Polybutylene Terephthalate is a large polymer and its surface would have no chemical reactivity, it was the consensus that this long-chain polymer would not penetrate the skin. Thus, the use of this ingredient is not of concern from a systemic toxicity point of view (CIR, 2014).

Reference :

CIR. Safety Assessment of Modified Terephthalate Polymers as Used in Cosmetics. 2014

Chemical Substance: POLYETHYLENE TEREPHTHALATE

EU INCI NAME:POLYETHYLENE TEREPHTHALATE

CAS: 25038-59-9

EINECS polymer

Function: film formers

Cosmetic Regulatory Summary:

EU Cosmetics Status: Not controlled

Saudi Cosmetics Status: Not controlled

US Cosmetics Status: Not controlled

Canadian Cosmetics Status: Not controlled

Regulatory Summary:

EU Classification: Unclassified

GHS Classification: Unclassified

REACH Annex XVII controlled: Not Controlled

REACH SVHC Candidate List: Not Controlled

Systemic Exposure Dosage / Margin of Safety:

SED Adult mg/kg bw/day: 6.412 No NOAEL Available

SED Child mg/kg bw/day: 23.038 No NOAEL Available

SED Baby mg/kg bw/day: 65.210 No NOAEL Available

NO(A)EL mg/kg bw day: -

Toxicological Summary:

Cosmetic Functions : Film Forming. Inert fine particles which reflect the light. Polyethylene Terephthalate is safe for use in food packaging that may be stored, heated, or microwaved. Considered as GRAS by FDA. In use tests of several different eye products containing PET up to 46.272%, the products were found to be non irritating and non sensitizing including in subject wearing contact lenses. An eyeliner containing Polyethylene Terephthalate (1.5%) was not irritating or sensitizing in a repeated insult patch test. In several tests of water stored in PET, the water was not mutagenic to *S. typhimurium* except for one test where the water was mutagenic after storage for 1 month but not at 3 and 6 months (CIR, 2014). CIR expert panel concluded that it is safe in present practices of use and concentration (up to 100% in leave on cosmetic product).

Reference:

Safety Assessment of Modified Terephthalate Polymers as Used in Cosmetic, CIR, 2014

EU INCI NAME: ACRYLATES COPOLYMER

CAS: 25133-97-5 / 25035-69-2 / 25212-88-8 / 25086-15-1 / 27322-15-2 / 9010-88-2

Function: Film former

EINECS polymer

Melting Point: 99.9°C

Appearance: Solid

Boiling Point: N/A

Log Kow: Not available

Vapour Pressure: 18.4 mmHg (20°C)

Water Solubility: Miscible

Cosmetic Regulatory Summary:

EU Cosmetics Status: Not controlled
Saudi Cosmetics Status: Not controlled by Saudi legislation
US Cosmetics Status: Not controlled
Canadian Cosmetics Status: Not controlled

Regulatory Summary:

EU Classification: Unclassified
GHS Classification: Unclassified
REACH Annex XVII controlled: Not Controlled
REACH SVHC Candidate List: Not Controlled

Systemic Exposure Dosage / Margin of Safety:

SED Adult mg/kg bw/day: 0.023	MoS - Adult 60kg: 10656.4	NO(A)EL mg/kg bw day: 250	
SED Child mg/kg bw/day: 0.084	MoS - Child 16.7kg: 2966.0	NOAEL test method:	Based on Acrylic acid in the drinking water given daily to rats in a 2-generation study IARC (1999)
SED Baby mg/kg bw/day: 0.238	MoS - Baby 5.9kg: 1047.8		

Toxicological Summary:

Acrylates Copolymer is a copolymer of two or more monomers consisting of acrylic acid, methacrylic acid or one of their simple esters. They are produced from primary polymer particles of about 0.2 to 6.0 micron average diameter. The flocculated agglomerates cannot be broken into the ultimate particles when produced. Acrylates copolymers are used in many types of cosmetic products and have been evaluated by the Cosmetic Ingredient Review (CIR) Expert Panel which concluded that the acrylates copolymers are safe for use in cosmetics when formulated to avoid irritation of the skin. Acrylates copolymer is expected to be poorly absorbed and so has limited systemic bioavailability. The primary toxicological concern with acrylates copolymer is with the unreacted monomers and/or residual chemicals such as solvents, plasticisers or catalysts. Little residual monomers are generally present in acrylates copolymer; although residual acrylic acid may be as high as 1500 ppm, typical levels are reported to be 10 to 1000 ppm (CIR, 2002). The residual concentration of monomers in Acrylates Copolymers have been reported. According to the evaluation of reports "Ten companies representing the majority of the production of polymers sold for cosmetic use indicated that residual acrylic acid concentrations in polymers are typically between 10 and 1000 ppm, with an upper limit of 1500 ppm. One source reported Acrylates Copolymer can contain residual amounts of <=20 ppm ethyl acrylate, methyl methacrylate, methacrylic acid, and acrylic acid; another source reported that three samples analyzed using GC contained <0.2 to 0.8 ppm acrylic acid, 0.8 to 2.6 ppm methyl methacrylate, and 1.3 to 3.9 ppm ethylene glycol dimethacrylate. Additionally, it was reported to CIR that two polymers, both defined as Acrylates Copolymer, contained different residual monomers; the first contained 36, 20, and 45 ppm n-butyl acrylate, methyl methacrylate, and methacrylic acid, respectively, and the second contained 1500 and 200 ppm stearyl acrylate and methacrylic acid, respectively" (Final report on the safety assessment of Acrylates Copolymer and 33 related ingredients (Int J Toxicol Vol:21, Suppl 3 (2002) pp 1-50). The published toxicokinetics data for acrylic acid and methyl acrylate suggest that both may be absorbed through the skin, the gastrointestinal tract and nasal cavity.

Acute toxicity: Acrylates copolymer (~24% or 30% solids) has low acute toxicity with reported LD50 values of > 2 to 9 g/kg (rat) after oral dosing, and > 16g/kg (rabbit) following dermal application. The LC50 > 5.2 mg/L (rat, 4-hr) on inhalation exposure to the aerosol (MMAD: 1.4µm). However, acrylic acid has shown moderate toxicity on dermal application (LD50 – 295 to 950 mg/kg; rabbit). **Irritation:** The skin irritation potential of the polymer has been evaluated undiluted in rabbits according to international OECD Guidelines No. 404. The test material (0.5g of dry polymer, moistened with 0.5 ml distilled water) was applied to the intact skin and held in place under a semi-occlusive binder for an exposure period of four hours. Following the exposure period, the binder was removed, and the remaining test article was rinsed off the skin with the test sites subsequently examined and scored for dermal irritation for up to seven days following patch removal. Under the test conditions, the test material caused mild skin irritation (Primary Irritation Index 0.3). Likewise an eye irritation test with the polymer has been conducted. Undiluted and 5% solution was instilled in the conjunctival sac of rabbits and left unrinsed. The undiluted test material produced moderate corneal and conjunctival irritation (maximum mean score 37.7 out of 110; class 5 on a 1 to 8 scale) which was resolved within 21 days whilst at a concentration of 5%, minimal eye irritation was demonstrated (conjunctival; maximum mean score 9.3 out of 110; class 3 on a 1 to 8 scale) (Lubrizol, 2004). Thus, it is concluded that the acrylate copolymer has minimal skin and minimal to moderate eye irritation.

Sensitisation: Acrylates copolymers did not produce a skin sensitisation reaction in series of guinea pig maximization test at challenge concentrations ranging from 5% to 75% following induction treatment with 0.5% to 25%.

Mutagenicity/Genotoxicity: Acrylates copolymer was non mutagenic or shown any evidence of genotoxicity in Ames tests, mouse lymphoma assays, in vitro chromosome aberration tests and in vivo bone marrow micronucleus test (CIR, 2002 and references cited therein).

Repeat dose toxicity: No repeated dose toxicity data is available for the acrylate copolymer. Surrogate data with other structurally similar acrylic acid polymers are available. Oral and dietary administration studies with carbomers reported significant bodyweight reductions at high doses (5000 mg/kg/day) administered for 30 days. However, no effect on bodyweight gain was seen following administration of a carbomer in rat at 5% in diet (corresponding to ~ 4000 mg/kg/day) for 21 days; and in a 90-day study at 1% in diet (~ 800 mg/kg bw/d). In the dog, a no-observable-adverse-effect level (NOAEL) of ~ 250 mg/kg bw/d (i.e. 1% in diet) was reported. Acrylates copolymer (polymer backbone was n-butyl acrylate, methyl methacrylate, methacrylic acid) formulation containing 69% ethanol was tested for inhalation toxicity in rats exposed whole-body for 13 weeks to 1, 10, or 30 mg/m3. The actual concentrations of polymer that the animals were exposed to were 0.185, 1.67, and 4.94 mg/m3 (CIR 2002 and references cited therein). No significant exposure related lesions or changes were noted and the formulation was concluded as having a relatively low pulmonary toxicity with the no-observable-adverse-effect level (NOAEL) of 10 mg/m3 of the formulation (corresponding to 1.67 mg/m3 acrylate copolymer). Acrylic acid monomer had no adverse effects on rats in a 12-month drinking water study at daily mean intake of 9, 61, 140, and 331 mg/kg bw/d; respectively. Sodium polyacrylate was concluded not to show teratogenicity or embryotoxicity in the rat at oral doses of up to 1125 mg/kg bw/d (high molecular weight material, 90000 Da) or 3000 mg/kg bw/d (low molecular weight material, 4500 Da). Maternal toxicity (3 deaths) was noted at 1125 mg/kg bw/d (high molecular weight sodium polyacrylate). Repeated dermal application in the mouse for 13 weeks using 0.1 ml volumes of acrylic acid monomer reported 1% concentration to be well-tolerated (McLaughlin et al. 1995 as cited in CIR, 2002). Acrylic acid administered at dose levels of 83, 250, or 750 mg/kg acrylic acid in the drinking water daily to rats in a 2-generation study reported a NOAEL of 250 mg/kg. IARC (1999) concluded that "acrylic acid is not classifiable as to its carcinogenicity to humans". "Read across" to data on carbomer provide the lowest NOAEL of 250 mg/kg bw/d.

Photo-induced toxicity: No data available indicating phototoxicity potential.

Other data (human): Human repeated insult patch test (HRIPT) has been conducted with the polymer material. 150 mg of a 10% distilled water paste of the polymer material was applied to the skin of 113 human volunteers and the skin irritation and sensitisation potential was assessed. The test material did not produce any evidence of skin irritation or skin sensitization under the conditions of the test (Lubrizol, 2004).

Dermal / percutaneous absorption: No data on dermal absorption is available for the copolymer. However, absorption of the monomer, acrylic acid, has been reported to be 24% of dose following application to rat skin samples in vitro with approximately 60% of the dose evaporating on application (Black et al. 1995). Hence, the dermal absorption of acrylates copolymer is expected to be less than 24% of the applied dose.

Margin of Safety: NOAEL: 250 mg/kg bw/d; Dermal Absorption (Dap) = 24 %

References:

Lubrizol 2004: Carbopol Ultrez 20 Polymer Toxicology studies. Noveon Consumer Specialties Toxicology and Microbiology studies; TOX-080, February 5, 2004. Available at: <http://www.lubrizol.com/PersonalCare/Products/Carbopol/CarbopolUltrez20.html>

CIR 2002: Final report on the safety assessment of acrylates copolymer and 33 related cosmetic ingredients. International Journal of Toxicology, 21 (Suppl. 3): 1-50, 2002. Accessed at: www.cir-safety.org

Chemical Substance: EPOXY RESIN

CAS: 61788-97-4

EINECS polymer

Regulatory Summary:

EU Classification: Unclassified
GHS Classification: Unclassified
REACH Annex XVII controlled: Not Controlled
REACH SVHC Candidate List: Not Controlled

Systemic Exposure Dosage / Margin of Safety:

SED Adult mg/kg bw/day: 0.097	No NOAEL Available
SED Child mg/kg bw/day: 0.351	No NOAEL Available
SED Baby mg/kg bw/day: 0.994	No NOAEL Available

Toxicological Summary:

Reaction product of epichlorohydrin and bisphenol-A. Also known as phenolic epoxy resin. When fully cured, would not be expected to cause irritation or allergy. Inherently withstands chemical attacks such as corrosion and stable at high temperature. Characteristically non-flammable, non-absorbent and durable.

EU INCI NAME:CI 77000

CAS: 7429-90-5

EINECS 231-072-3

Function: Colour

Appearance: solid

Melting Point: Not available

Log Kow: Not available

Boiling Point: Not available

Water Solubility: Not available

Vapour Pressure: Not applicable

Cosmetic Regulatory Summary:

EU Cosmetics Status: Approved colour all products

Saudi Cosmetics Status: Permitted colour field 1 All products

US Cosmetics Status: Except lip and oral products. All product including eye area 73.2645

Regulatory Summary:

EU Classification: R15-17

REACH Annex XVII controlled: Not Controlled

REACH SVHC Candidate List: Not Controlled

Systemic Exposure Dosage / Margin of Safety:

SED Adult mg/kg bw/day: 0.000 MoS - Adult 60kg: 767263.4

NO(A)EL mg/kg bw day: 30

SED Child mg/kg bw/day: 0.000 MoS - Child 16.7kg: 213554.9

NOAEL test method:

PTWI of 2 mg/kg bw for aluminium was established based on NOAEL 30 mg/kg/d at the 74th JECFA (2011)

SED Baby mg/kg bw/day: 0.000 MoS - Baby 5.9kg: 75447.5

Toxicological Summary:

Aluminium is the most abundant metal in the earth crust and not classified as toxic. Aluminium powder form is pyrophoric and can cause an explosive reaction in air. It reacts to form oxides and hydroxides which are form alkaline solutions with water. Although in powder form aluminium can react with water, this is unlikely to occur in an oil-based cosmetic formulation. Hence incorporation into an oil-based cosmetic formulation is unlikely to give rise to skin or eye irritation. Highly reflective particles which give colour by reflectance. The product is essentially inert. May cause a foreign body reaction should they enter the eye and enhance the eye irritancy potential of other components in a product. If inhaled may cause slight transient respiratory irritation. Animal testing shows long term exposure to Aluminium Oxides may cause lung disease and cancer, depending on the size of the particle. The smaller the size, the greater the tendencies of causing harm. From the IARC Monograph, [Vol 34, pp 37 (1984); Suppl. No 7, pp 89 (1987)], Aluminum is classified in Group 1 as Carcinogenic to Humans (updated in 2003).

Aluminum is an approved colour permitted for use in cosmetic products in the EU and also the US (73.26454).

For the use in the United States of America, Aluminum is exempt from batch certification and is registered at the FDA under the following uses:

- FDA - Color Additives Conditionally Approved for Use in Cosmetics 21 CFR 73.2645.
- FDA - Color Additives Conditionally Approved for Use in Drugs 21 CFR 73.1645.
- FDA - Indirect Food Additives 21 CFR 175.105.

Aluminum is listed at the annex IV part 1 of the Cosmetics Directive and is allowed in all cosmetic products in EU and is approved by the FDA for use in external applied drugs and cosmetics. ALUMINUM POWDER toxicity: Low acute oral toxicity (DL50=261mg/kg rat). Not genotoxic (Ames 10mM, Syrian hamster embryo cells), very low potential for skin irritation (largely used in antiperspirant), very rare cases of allergy reported. No case of neurotoxicity or osteotoxicity reported in man. Transcutaneous permeation is 0.02%. ATSDR set a MRL value for exposure by oral route to Aluminum derived from a LOAEL of 100 mg/kg /d for neurological effects in mice exposed to aluminum lactate in the diet during gestation, lactation, and postnatally until 2 years of age (Golub et al, 2000). This MRL value is 1 mg Al/kg bw/d (ATSDR-2008).

The average human intake of aluminum (CI 77000) is estimated to be 25 mg/day. This intake comes from food and drinking water. The intake from pharmaceuticals, anti-acids and some buffered analgesics can range from 10- 5000 mg/day. Evaluation performed by the Joint FAO/WHO Expert Committee on food additives concludes to a tolerable intake of 7 mg/kg body weight for aluminum powder but no ADI was allocated. Taking into account that ingredient alumina or aluminum oxide is considered as inert and insoluble in water and therefore its bioavailability is extremely low, no systemic effects have to be feared.

JECFA 2011 PTWI for aluminium was 2 mg/kg based on a no-observed-adverse-effect level (NOAEL) of 30 mg/kg body weight per day and application of a safety factor of 100. The estimated percentages of absorbed aluminium were 0.6% for normal skin and 10.7% for stripped skin (Risk assessment of the exposure to aluminium through food and the use of cosmetic products in the Norwegian population, 11-504_final, 2013)

Chemical Substance: CI 17200 (D&C RED NO. 33)

EU INCI NAME:ci 17200

CAS: 3567-66-6

EINECS 222-656-9

Function: Colour

Appearance: Dark red/brown powder

Melting Point: Not available.

Log Kow: Log Pow: ≤ 0.5 (extrapolated)

Boiling Point: N/A

Water Solubility: >2.5%

Vapour Pressure: N/A

Cosmetic Regulatory Summary:

EU Cosmetics Status: Approved colour all products

Saudi Cosmetics Status: Permitted colour field 1 All products

US Cosmetics Status: All cosmetic products 74.2333 Lip products (3% max) Mouthwashes/dentifrices (GMP). Not permitted in eye area

Regulatory Summary:

EU Classification: unclassified

GHS Classification: unclassified

REACH Annex XVII controlled: Not Controlled

REACH SVHC Candidate List: Not Controlled

Systemic Exposure Dosage / Margin of Safety:

SED Adult mg/kg bw/day: 0.000 MoS - Adult 60kg: 27402.2

NO(A)EL mg/kg bw day: 25

SED Child mg/kg bw/day: 0.003 MoS - Child 16.7kg: 7626.9

NOAEL test method:

multi-generation, repeated oral dose study in rat (SCCP, 2007)

SED Baby mg/kg bw/day: 0.009 MoS - Baby 5.9kg: 2694.5

Toxicological Summary:

A colour widely used with a good history of safe use. An FDA colour widely used as a food additive and subjected to extensive investigations. Unlikely to cause adverse effects at the typical concentrations used in cosmetics.

The Cosmetics Directive 2009/36/EC has amended the use of this colour to be used in non oxidative hair dyes at a maximum concentration of 0.5%. This does not effect its use in other permitted cosmetic product types.

Permitted in US in all cosmetic products 74.2333 Lip products (3% max)

Acute oral toxicity (LD50) was > 3 160 mg/kg bw in rats and > 1 000 mg/kg bw in dogs.

In a multi-generation study in rats, the NOAEL was set at 25 mg/kg bw/d.

Skin sensitization: Considered not to be irritant to rabbit skin and eye. It was found not to be a skin sensitiser in a Guinea pig maximisation test.

Dermal absorption: in vitro, the A max of 15.2 µg/cm² is reported. In an in vitro skin absorption study (OECD 428), the percentages of penetration are 0.9% and 0.12% of the applied dose. As two few chambers were used in the study, the default rate of skin penetration was considered to be 1 %.

Genotoxicity: Did not induce gene mutations in the gene mutation assay neither in bacteria

nor in mammalian cells at the tk locus of mouse lymphoma cells. Can be considered to have no relevant mutagenic potential in vivo. Carcinogenicity: Non carcinogenic after 2 year study in vivo

Reference: SCCP/1102/07

EU INCI NAME: CI 16035

CAS: 25956-17-6

EINECS 247-368-0

Function: Colour

Appearance: powder

Melting Point: 349.8

Log Kow: -0.55

Water Solubility: 4.2

Cosmetic Regulatory Summary:

EU Cosmetics Status: Approved colour all products
Saudi Cosmetics Status: Permitted colour field 1 All products
US Cosmetics Status: Generally, including cosmetics 74.2340 including the area of the eye

Regulatory Summary:

EU Classification: Unclassified
GHS Classification: Unclassified
REACH Annex XVII controlled: Not Controlled
REACH SVHC Candidate List: Not Controlled

Systemic Exposure Dosage / Margin of Safety:

SED Adult mg/kg bw/day: 0.130	MoS - Adult 60kg: 5332.4	NO(A)EL mg/kg bw day: 695	
SED Child mg/kg bw/day: 0.468	MoS - Child 16.7kg: 1484.2	NOAEL test method:	reproductive study in rats
SED Baby mg/kg bw/day: 1.325	MoS - Baby 5.9kg: 524.3		

Toxicological Summary:

A thoroughly investigated dye with low oral toxicity. Widely used in food (an FDA colour) and cosmetics. Unlikely to cause adverse effects when used in cosmetics. The Cosmetics Regulation 1223/2009 has amended the use of this colour to be used in non oxidative hair dyes at a maximum concentration of 0.4%. This does not effect its use in other permitted cosmetic product types.

EFSA have reviewed Allura Red in its report published in 2009.

Toxicological findings: Acute Toxicity LD50 10000 (rat) and 5000 (dog) mg/kg bw, respectively

Allura Red has an ADI of 7 mg/kg bw/day and was based on a NOAEL of 695 mg/kg bw/day derived from a reproductive toxicity study in rats which revealed slight growth suppression observed mainly at the high test levels of 2595 mg/kg bw/day in F1 and F2 pups.

Scientific Opinion on the re-evaluation of Allura Red AC (E 129) as a food additive 1, EFSA Journal 2009; 7(11):1327

Chemical Substance: CI 42090 (FD&C BLUE NO. 1)

EU INCI NAME: CI 42090

CAS: 3844-45-9

EINECS 223-339-8

Function: Colour

Appearance: solid

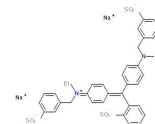
Melting Point: Not available

Log Kow: Not available

Boiling Point: N/A

Water Solubility: A (µg/cm²) = 35.5 (µg/cm²). [SCCNFP, 2004]

Vapour Pressure: N/A



Chemical Structure:

Cosmetic Regulatory Summary:

EU Cosmetics Status: Approved colour all products
Saudi Cosmetics Status: Permitted colour field 1 All products
US Cosmetics Status: Generally, including cosmetics 74.101.

Canadian Cosmetics Status: Approved colour all products including eyes

Regulatory Summary:

EU Classification: Unclassified
GHS Classification: Unclassified
REACH Annex XVII controlled: Not Controlled
REACH SVHC Candidate List: Not Controlled

Systemic Exposure Dosage / Margin of Safety:

SED Adult mg/kg bw/day: 0.000	MoS - Adult 60kg: 2685421.9	NO(A)EL mg/kg bw day: 630	
SED Child mg/kg bw/day: 0.000	MoS - Child 16.7kg: 747442.4	NOAEL test method:	female rat, one generation chronic oral study
SED Baby mg/kg bw/day: 0.002	MoS - Baby 5.9kg: 264066.4		

Toxicological Summary:

A dye thoroughly tested for use as a food additive (an FDA colour) and cosmetic dye. Unlikely to cause adverse effects when used in cosmetics. Permitted in US for use in all cosmetic products including those used near the eye. Unlikely to cause adverse effects at the typical concentrations used in cosmetics. Widely used and well accepted in cosmetic products. Canada cosmetic use; coal tar dye permitted in the area of the eye. The Cosmetics Directive 2009/36/EC has amended the use of this colour to be used in non oxidative hair dyes at a maximum concentration of 0.5%. This does not effect its use in other permitted cosmetic product types. Approved colour for use in finger paints EN71:7

CI 42090 (Acid Blue 9; CAS No. 3844-45-9) is also allowed as a food colorant in the EU with an ADI of 1-10 mg/kg bw/day. Acid Blue 9 has long history of use in a wide range of applications and it is "generally recognized as safe (GRAS)". Has low acute toxicity (Oral LD50 > 2000 mg/kg/d; rat, mouse). No data by the dermal and inhalation routes of exposure. Moderate skin irritation was reported in 40 of 207 volunteers in a testing of 5% aqueous solution but at 1%, it was non-irritating to skin in mice and rabbits. 10% aqueous solution of CI 42090 produced transient eye irritation in rabbits treated over a 4-week period and in HET-CAM test, only slight irritation was observed. Also, no irritation was reported in the neutral red uptake (NRU) assay on human keratinocytes at doses from 0.68 -10 mg/ml; the NRU-50 is reported > 10 mg/ml. In a local lymph node assay (LLNA), CI 42090 was found non-sensitising on dermal contact (stimulation indices < 3) and showed no potential for photosensitisation. Not mutagenic or genotoxic (bacterial, chromosome aberration, in vivo UDS assays). No significant toxicity effect was reported in oral administration of Acid Blue 9 at level of 4% in the diet (~ 2000 mg/kg bw/day) for 3 weeks in rats. Chronic toxicity studies have been conducted in rats and mice with Acid Blue 9. Mice were less sensitive with no adverse effects reported at the highest concentration of 9000 mg/kg bw/day (5% in diet). However, decreased in body weight and survival was seen in female rats dosed 2000 mg/kg bw/day (~ 2% in diet) and the NOAEL was concluded as 630 mg/kg bw/day (1% Acid Blue 9 in diet). The weight of evidence from available data indicates that it is not carcinogenic on oral administration. Acid Blue 9 was not teratogenic in rats and rabbits in doses up to 2000 mg/kg bw/day and 200 mg/kg bw/day, respectively. Reproductive toxicity was not reported with Acid Blue 9 in one-generation study (NOAEL: 1000 mg/kg bw/day; 2% in the diet). However, effects on offspring (bodyweights) and slight increase in the incidence of lung and kidney abnormalities in adult males were reported in a two-generation study and the LOAEL was determined to be 1000 mg/kg bw/day. The oral and percutaneous absorption of Acid Blue 9 is low (< 5%). An in vitro study on pig skin revealed maximum penetration rate of 35.5 µg/cm² within 24 hours (i.e. 0.60% of the applied dose) for a 0.5% product.

Note: In the absence of NO(A)EL data, the Margin of Safety (MoS) has not been calculated. Unless otherwise determined and in the absence of literature or other experimental data, a Dermal Absorption (DAP) of 100% is taken as the worst case scenario. NO(A)EL: No Observed Adverse Effect Level; MoS: Margin of Safety; SED Systemic Exposure Dosage; Calculation of Margin of Safety: $MoS = NO(A)EL / SED$

Reference for skin surface area, exposures and application quantities are derived from:

1. RIVM Report 320104001/2006
2. References cited in Dermal Sensitization Quantitative Risk Assessment (QRA) For Fragrance Ingredients, 2006 revision
3. Exposure factors handbook 2011
4. SCCS's Notes of Guidance for the Testing of Cosmetic Ingredients and their Safety Evaluation 8th Revision SCCS/1501/12
5. Colipa Data SCCNFP/0321/02
6. McNamara et al, Food Chem. Tox; 2007, 45, 2086; Loretz et al, Food Chem. Tox; 2008, 46, 1516
7. Body weights taken from Exposure factors handbook 2009 Update and mean values have been used unless specified otherwise
8. ConsExpo database
9. New default values for the spray model, RIVM, March 2010
10. Commission Implementing Decision of 25 November 2013 on Guidelines on Annex I to Regulation (EC) No 1223/2009 of the European Parliament and of the Council on cosmetic products (2013/674/EU)

This formulation has been assessed by Intertek in accordance with Part B, Annex I to Regulation (EC) No. 1223/2009 of the European Parliament and of the Council of 30 November 2009 on cosmetic products. The safety assessment is based upon the chemical specification and toxicological profile of the ingredients as supplied at the time of assessment and an assessment of the final cosmetic product. The supplier to this safety assessment is advised to ask for a new safety evaluation if any change in formulation occurs, change in raw materials used, abnormally high number of adverse events are recorded, changes in recommended uses or other circumstances that may affect the safety of this product.

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